

POLYCHRONIS PATTAKOS, PHD

AI Engineer | LLM Applications & Agentic Systems • Applied ML • RAG • Data Analysis • Full-Stack

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PROFILE

Electrical & Computer Engineer (PhD) with hands-on experience building and deploying LLM-powered applications — agentic tool-calling, retrieval-augmented generation (RAG), and full-stack API integration — delivered end-to-end from backend architecture to live production deployment. Combines this applied AI engineering work with hands-on applied machine learning and data analytics (time-series anomaly detection, sensor fusion, explainable risk-scoring models, built from first principles when library constraints required it) and deep domain expertise in sensor systems and applied research, a research-grade ability to reason rigorously about data, and a track record of coordinating complex technical projects across engineering teams.

SELECTED AI ENGINEERING PROJECTS

Project Intelligence Platform — AI-Powered Facility Security Risk Assessment System

Personal project, 2026 · extends professional security consulting experience into an end-to-end AI platform •

github.com/drPolPat/project-intelligence-platform • [Live demo](#)

- Built a full-stack AI platform combining ML risk prediction, RAG-grounded standards Q&A, and an agentic assistant orchestrating both — six independently-verified phases (data → ML → RAG → agent → API → dashboard), deployed end-to-end (FastAPI on Railway, React on Vercel), using entirely synthetic data.
- Trained and validated two risk-prediction models via 10-split cross-validated robustness checks — an initial single-split result favoring Random Forest reversed under the full check; what held was a specific, mechanistically explained finding: Random Forest alone detects a nonlinear "compounding risk" pattern structurally invisible to linear models (0.57 vs. 0.00 recall on Critical-tier cases, 10/10 splits).
- Built a citation-grounded RAG system with two independent refusal mechanisms, proving via a hard-negative test that similarity-based retrieval alone was insufficient — a topically-adjacent but out-of-scope query scored well above the retrieval threshold, and only the model's own grounding discipline caught it.
- Designed a tool-calling agent (Claude API) composing the ML and RAG systems under a zero-tolerance grounding constraint — caught a subtle violation (agent appending unauthorized synthesis after a verbatim tool relay) that a standard tool-selection eval couldn't detect, and fixed it with a stricter exact-match regression check.
- Exposed the platform's tools via a FastMCP server as an independently deployable service, reusing existing tool schemas/descriptions verbatim rather than redefining them — deliberately isolated its dependencies after discovering the MCP framework's own upgrade path would have broken the deployed backend's pinned FastAPI version; verified via both an in-memory client and the real MCP protocol (stdio subprocess), matching every prior result.
- Added CI/CD via GitHub Actions gated on the project's own eval suites (RAG retrieval/citation checks, agent tool-selection) — the first real run surfaced two genuine pre-existing bugs invisible in local development (a JSON-serialization crash in the agent eval, an undocumented gap between the claimed and actual RAG baseline), both root-caused and fixed rather than papered over with a looser threshold.
- Tech: Python, FastAPI, React, Anthropic Claude API, scikit-learn, ChromaDB, fastembed, MCP (FastMCP), agentic tool-calling, Docker, Railway, Vercel.

SteelSense AI — LLM-Powered Structural Health Monitoring Assistant

Personal project, 2026 · extends doctoral research into an AI application • github.com/drPolPat/steelsense-ai • [Live demo](#)

- Built an end-to-end agentic AI application applying LLM-based reasoning to structural health monitoring, grounded in doctoral research on magnetic sensor-based stress measurement.
- Designed a tool-calling agent (Claude API) combining a physics-based anomaly detection pipeline (thermal correction, drift/spike/divergence classification) with a RAG knowledge base for grounded, citation-backed answers.
- Built and deployed full-stack: FastAPI backend + React dashboard, containerized and live on Railway and Vercel, with CORS, environment configuration, and API key management handled for a production-style deployment.
- Implemented and compared two agent orchestration approaches (manual tool-calling loop and LangGraph StateGraph) against the same eval set.
- Tech: Python, FastAPI, React, Anthropic Claude API, ChromaDB (RAG), agentic tool-calling, Vercel, Railway.

PhD Knowledge Assistant — Grounded RAG over Doctoral Research

Personal project, 2026 · extends doctoral thesis and peer-reviewed publications into a domain-expert QA system •

github.com/drPolPat/phd-knowledge-assistant • [Live demo](#)

- Built a citation-grounded RAG assistant over my own PhD thesis and IEEE/Micromachines publications on magnetic sensors for nondestructive steel testing, with strict source-grounding and explicit refusal on out-of-corpus questions.
- Designed a custom evaluation harness scoring four axes (retrieval accuracy, citation validity, LLM-judged faithfulness, refusal correctness) against a hand-verified 20-question reference set — achieved 20/20 after diagnosing and fixing a retrieval regression through systematic experimentation (chunking strategy, retrieval depth).
- Built and deployed full-stack: FastAPI backend + React frontend, containerized and live on Railway and Vercel, with a local vector index (no external DB dependency).
- Documented every experiment — including a reverted change — in the eval report, treating negative results as evidence rather than discarding them.
- Tech: Python, FastAPI, React, Anthropic Claude API, fastembed (local embeddings), custom eval harness (LLM-as-judge), Vercel, Railway.

Sensor Fusion & Anomaly Detection — Multi-Generation Sensor Simulation & Fatigue Detection

Personal project, 2026 · simulates and extends doctoral sensor research into an ML anomaly-detection system •

github.com/drPolPat/sensor-fusion-anomaly-detection • [Live demo](#)

- Simulated a 4-generation magnetic sensor network from my doctoral research (lab reference, electromagnetic yoke, single- and multi-Hall permanent-magnet sensors), with physically grounded noise models (Villari effect, 1/f drift, thermal coupling) validated against Welch PSD and Allan deviation diagnostics.
- Built a two-stage calibration and Kalman fusion pipeline correcting noisy field sensors against a sparse lab-reference signal, reducing fused RMSE from 4.3–11.2 MPa (raw sensors) to 1.6 MPa, with performance holding up between calibration campaigns.
- Implemented isolation forest and LSTM autoencoder anomaly detectors from scratch in NumPy (including a gradient-checked hand-written backprop-through-time) after PyTorch and scikit-learn were blocked by the target machine's security policy — validated detector stability via 5-seed cross-validation rather than reporting single-run results.
- Built a live React dashboard visualizing the simulated sensor network, fusion output, and real-time anomaly flags/metrics across configurable fatigue scenarios.
- Wrote a full model card documenting synthetic-data scope, known limitations (false-alarm-rate seed sensitivity, calibration coverage gap), and what real hardware validation would require.
- Tech: Python, NumPy/SciPy (hand-implemented ML — no PyTorch/sklearn), Kalman filtering, FastAPI, React, Railway, Vercel.

Facility Security Risk Analytics — Explainable ML for Physical Security Risk Assessment

Personal project, 2026 · applies professional security risk assessment methodology to a data analytics/ML pipeline •

github.com/drPolPat/facility-security-risk-analytics • [Live demo](#)

- Built a weighted multi-factor risk-scoring framework for physical security assessment across facility archetypes (cultural heritage, commercial, mixed-use, critical infrastructure), informed by professional risk consulting work on infrastructure security projects.
- Trained and compared two regression models (Ridge, Gradient Boosting) predicting incident likelihood from risk factors, explicitly designing a threat-exposure × countermeasure interaction effect to test whether the models could recover a known nonlinear relationship — validated via SHAP feature importance analysis.
- Ran an empirical sample-size stability check (n=52 vs. n=300), demonstrating how SHAP rankings on shallow tree models can be unstable on small samples and documenting the methodology transparently.
- Built and deployed full-stack: FastAPI backend with an interactive what-if endpoint (real-time model re-scoring on countermeasure changes) + React dashboard with radar charts and SHAP visualization, containerized and live on Railway and Vercel.
- Tech: Python, scikit-learn (Ridge, Gradient Boosting), SHAP, pandas, FastAPI, React, Railway, Vercel.

ScholarCore AI

Client project via Reticulab AI LTD, 2025 (see Work Experience)

- Full-stack AI engineering on an early-stage startup product: built and integrated APIs, data ingestion/transformation pipelines, and lightweight ML workflows within the product's core system architecture.

Contributed to technical design decisions and roadmap planning alongside prototyping and validation of new AI-driven features.

SKILLS

AI / LLM Engineering: LLM integration (Anthropic Claude API), Retrieval-Augmented Generation (RAG), agentic tool-calling, LangGraph, ChromaDB, MCP (FastMCP), prompt design & evaluation, Anthropic Academy: AI Fluency (completed)

Software Engineering: Python, FastAPI, JavaScript, React, HTML/CSS, REST APIs, Git/GitHub, Vercel, Railway

Machine Learning: Isolation forest, LSTM autoencoders, Kalman filtering, time-series anomaly detection (implemented from first principles in NumPy — sensor-fusion-anomaly-detection); regression modeling & explainability (Ridge, Gradient Boosting, SHAP — facility-security-risk-analytics); Coursera Machine Learning Specialization; Kaggle certificates (Python, Intro to ML, Pandas)

Embedded & Sensor Systems: Arduino, ESP32, Raspberry Pi (Python/Linux), Teensy, Altium Designer, Autodesk EAGLE, ANSYS Discovery AIM

Evaluation & Testing: Custom eval harnesses (retrieval accuracy, citation validity, LLM-as-judge faithfulness scoring, refusal correctness); statistical validation via multi-seed cross-validation; grounded/ citation-based response verification.

Deployment: Production deployment and environment/secrets management (Railway, Vercel, GitHub Actions); Docker (containerized backend, optimized cold-start by baking the embedding model into the image at build time).

Other: AutoCAD, Microsoft Office

WORK EXPERIENCE

Electrical and Computer Engineer / Full-Stack Engineer

Reticulab AI LTD. • Jul 2025 – Dec 2025 • London Area, UK • Remote

- Full-stack engineering for the AI startup project ScholarCore AI (see Selected Projects above) — designed and built frontend/backend systems and monitored system performance in production.

Electrical and Computer Engineer

Safe Business Consulting (SBC) LTD. • Mar 2026 – now • Athens, Greece

- Technical project management and coordination for large-scale infrastructure and security-systems projects, including risk assessment of critical infrastructure and multi-stakeholder technical communication.
- Selected projects: National Archaeological Museum expansion, The Ellinikon Mall, Riviera Tower, HEDNO substations/warehouses, HEDNO/Amazon Data Center.
- Developed a data-driven risk-scoring methodology (see Facility Security Risk Analytics project) applying this work's domain framework to a quantitative, explainable ML model.

College Professor / Instructor

New York College • Oct 2025 – now • Athens, Greece

- Teach Computer Science Fundamentals, Introduction to Artificial Intelligence, and Introduction to Robotics at BSc level.

Freelance Electrical & Software Engineer

Self-employed • May 2021 – now

- Related projects include IoT smart-systems development (Scirocco S.A. – ANYWAY Pay-As-You-Throw system; City Post S.A. – The Green City smart recycling bins) and safety engineering consulting (Municipality of Lykovrisi-Pefki).

Postdoctoral Researcher

NTUA / University of Ioannina / Mitis Kyvernochoros Technologias S.A. • Jul 2023 – Aug 2024

- R&D on the AMSA project: autonomous multifunctional sensors for measuring fatigue in metallic naval structures — the applied domain foundation for the SteelSense AI project above.

EDUCATION

PhD, Electrical and Computer Engineering

National Technical University of Athens • Oct 2019 – Jul 2023

- Thesis: Development of Magnetic Circuits for Use in Autonomous Sensors for Measuring Internal Stresses in Steels. Supervisor: Prof. Evangelos Hristoforou.

MSc, Robotics and Automation Engineering

National Technical University of Athens • Oct 2017 – Feb 2019 • Grade: 8.67/10

Integrated MSc, Electrical and Computer Engineer

Electrical, Electronics & Automation Systems • *National Technical University of Athens* • Jul 2006

SELECTED PUBLICATIONS

- P. Pattakos et al., "Development of an Autonomous Magnetic Permeability Sensor," IEEE Transactions on Magnetics, 2022.
- P. Pattakos et al., "Magnetic harvester for autonomous steel health monitoring system based on Hall effect measurements," Micromachines, 2023.

LANGUAGES

Greek (native) • English (C2, Cambridge Proficiency) • German (B1, Goethe Zertifikat) • French (basic)